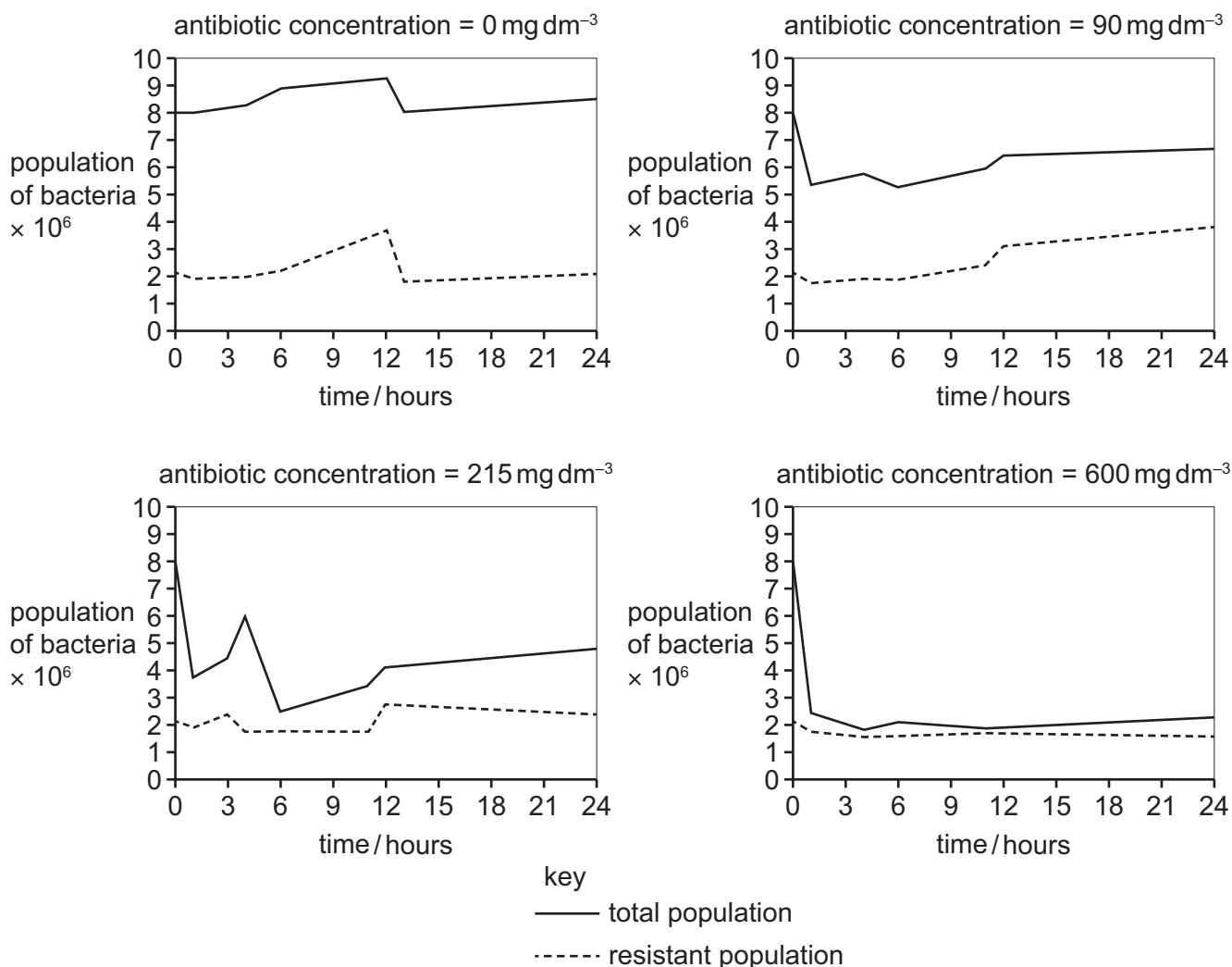


- 37** Scientists investigated the effect of increasing concentrations of an antibiotic on the development of antibiotic resistance in bacteria.

The scientists grew four groups of bacteria and added a different concentration of antibiotic to each group. The number of resistant bacteria and the total population of bacteria were measured at intervals for 24 hours for each group.

The graphs show the results.



Which statements are correct conclusions that can be made from the results of this investigation?

- 1 Increasing the concentration of antibiotic decreases the population of non-resistant bacteria at the end of 24 hours.
- 2 The proportion of antibiotic-resistant bacteria increases with increasing concentrations of antibiotics.
- 3 Increasing the concentration of antibiotic always increases the number of resistant bacteria.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only